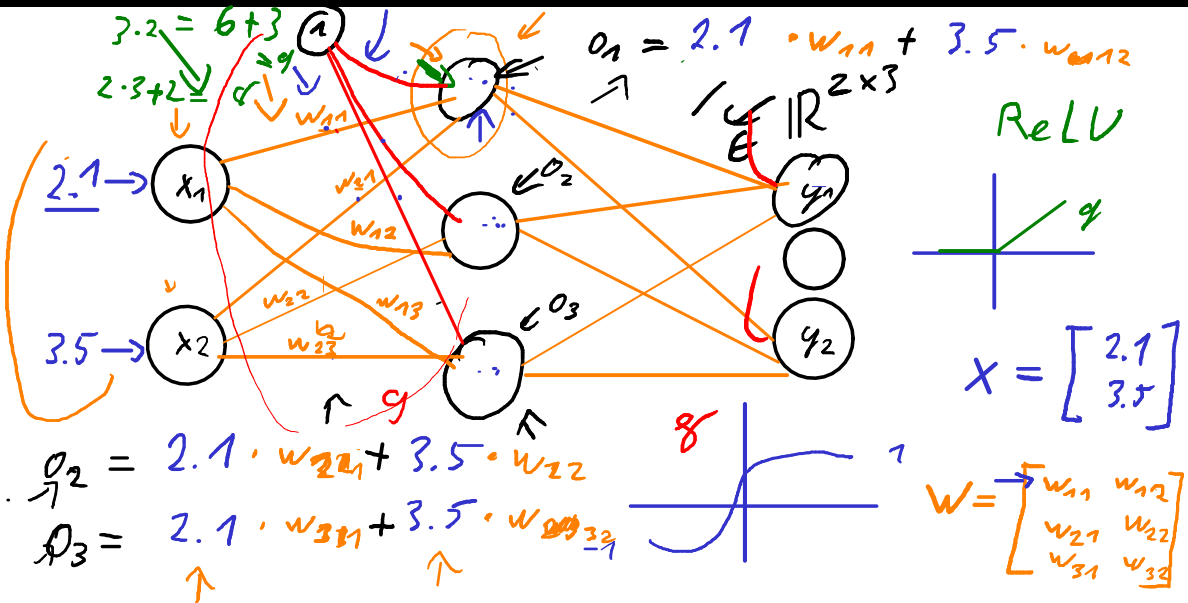


# Notes



# Notes

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2.1 \\ 3.5 \end{bmatrix}$$

$$W = \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \\ w_{31} & w_{32} \end{bmatrix} \leftarrow \begin{matrix} 2.1 & 3.5 \\ w_{11} & w_{12} \\ w_{21} & w_{22} \\ w_{31} & w_{32} \end{matrix}$$

$$W \cdot X$$



$$W \in \mathbb{R}^{3 \times \underline{2}}$$

$$X \in \mathbb{R}^{\underline{2}}$$

$$W \cdot X + b = \begin{bmatrix} x_1 \cdot w_{11} + x_2 \cdot w_{12} \\ x_1 \cdot w_{21} + x_2 \cdot w_{22} \\ x_1 \cdot w_{31} + x_2 \cdot w_{32} \end{bmatrix} + b = 0 = \begin{bmatrix} 0_1 \\ 0_2 \\ 0_3 \end{bmatrix}$$

$\uparrow$   $\leftarrow \in \mathbb{R}^3$

$$g(0) = \begin{bmatrix} g(0_1) \\ g(0_2) \\ g(0_3) \end{bmatrix} \in \mathbb{R}^3$$

# Notes

$$g(Wx + b) =$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T = [x_1 \ x_2]$$

$$x^T W = x W$$

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ x_{31} & x_{32} \end{bmatrix}$$

$$g(XW + b) =$$

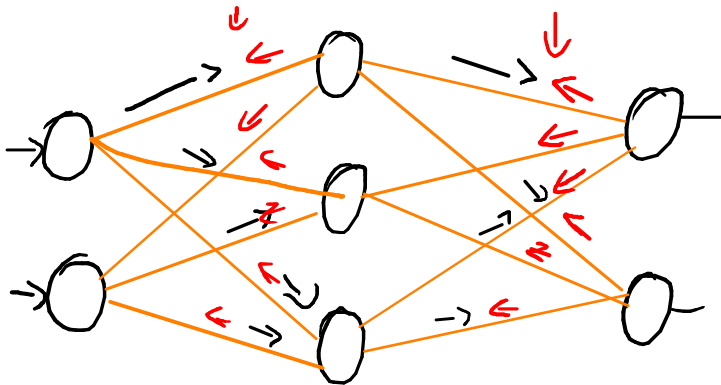
$$b \in \mathbb{R}^2$$

$$X \in \mathbb{R}^{m \times 2}$$

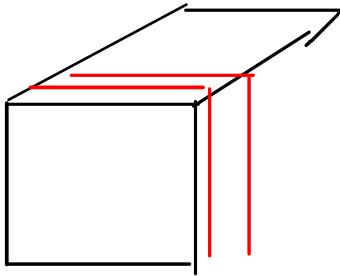
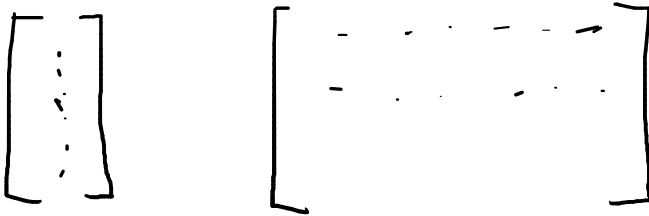
$$W \in \mathbb{R}^{2 \times 2}$$

$(B, \dots \rightarrow)$

## Notes



# Notes

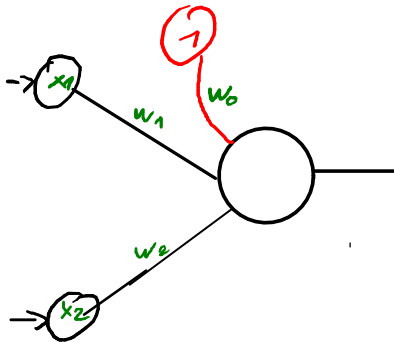


$$X \in \mathbb{R}^{10 \times 2} \quad W \in \mathbb{R}^{2 \times m}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x_{1,1} & x_{1,2} \\ x_{2,1} & x_{2,2} \\ x_{3,1} & x_{3,2} \\ \vdots & \\ x_{10,1} & x_{10,2} \end{bmatrix} \begin{bmatrix} \tilde{x}_{1,1} \\ \tilde{x}_{1,2} \\ \vdots \end{bmatrix} W$$

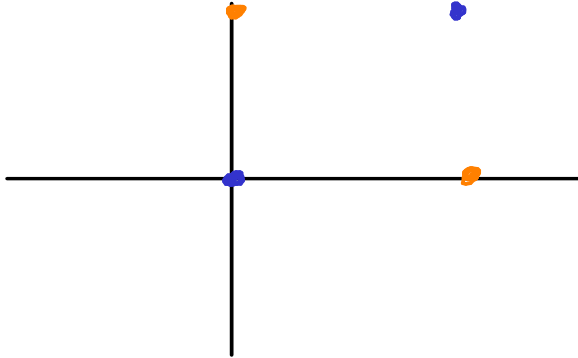
# Notes



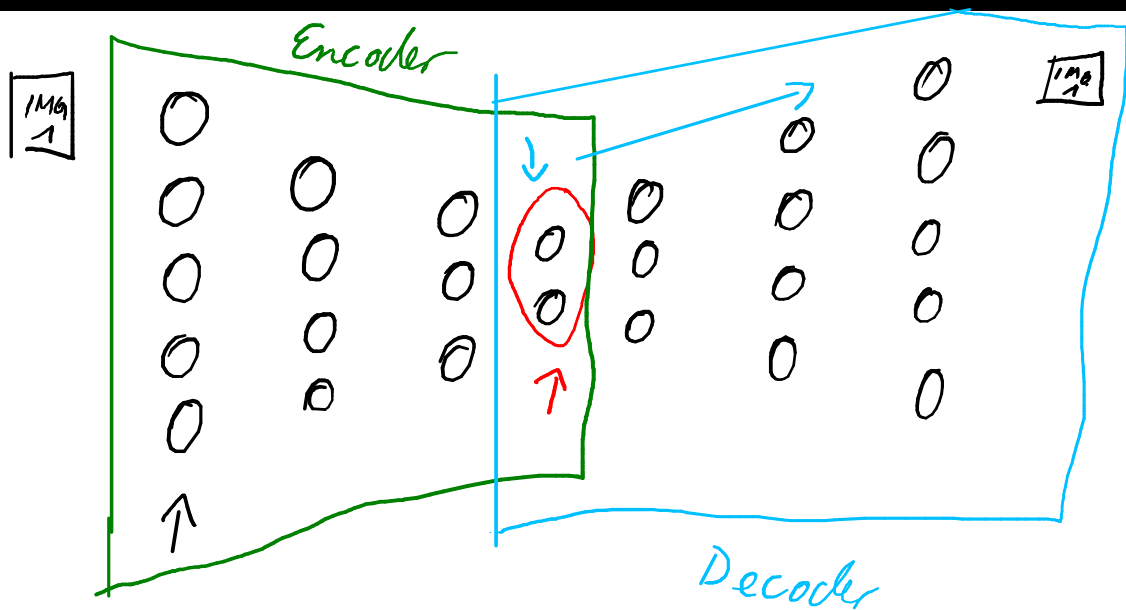
$$g(x) = \begin{cases} 1 & \text{if } w_0 + w_1 x_1 + w_2 x_2 > 0 \\ 0 & \text{else} \end{cases}$$



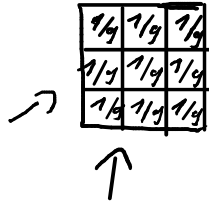
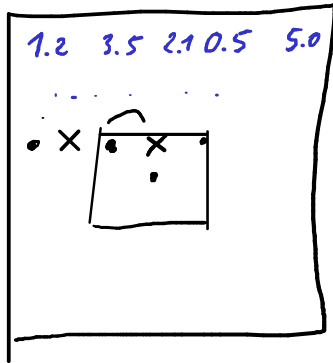
# Notes



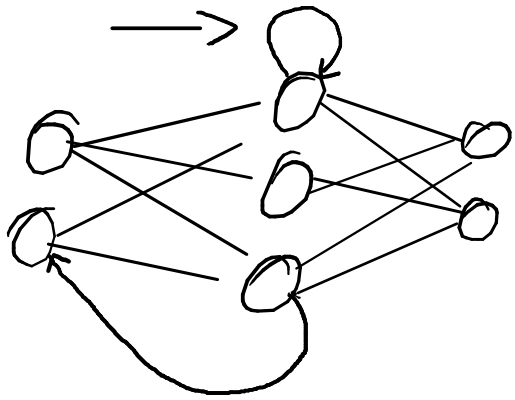
# Notes



# Notes



# Notes



# Notes

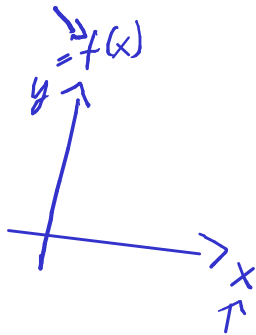
$$f(x) = 2x^2 + x - 5$$

$$f'(x) = 2 \cdot 2x + 1$$

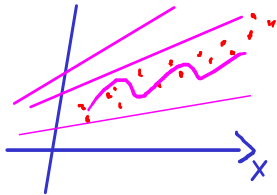
$$\{ (2, 5), (-1, 4) \dots$$

$\uparrow$  input  $x$

$\downarrow$  output  $y$



# Notes



$$f(x) = \underset{\uparrow}{w_1} \cdot x + \underset{\uparrow}{w_0}$$

$$f'(x) = \frac{df}{dx}$$

$$\frac{df}{dw_0} = 1$$

$$\frac{df}{dw_1} = x$$

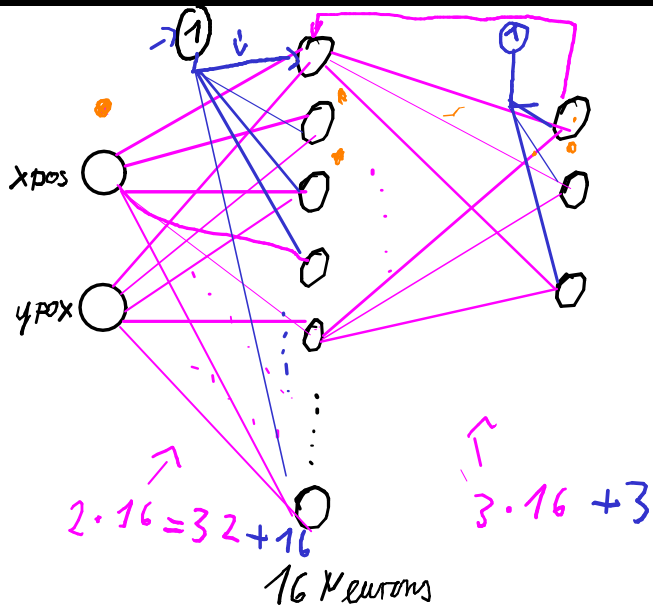
$$\begin{bmatrix} 1 \\ x \end{bmatrix}$$

↑

# Notes

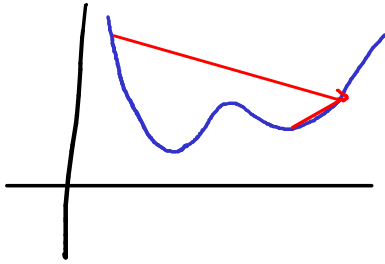


# Notes

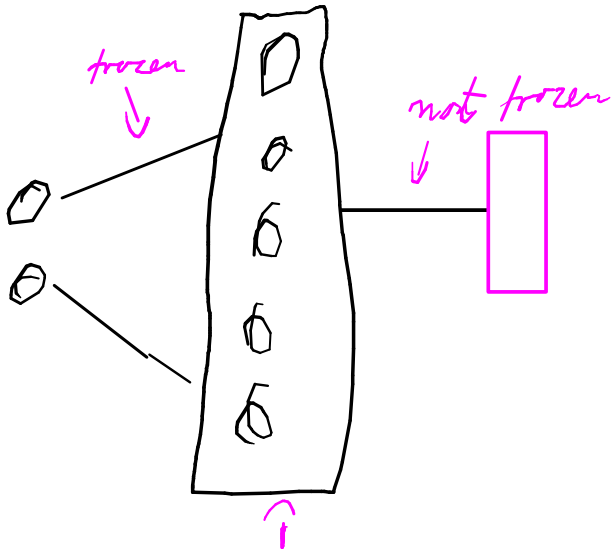




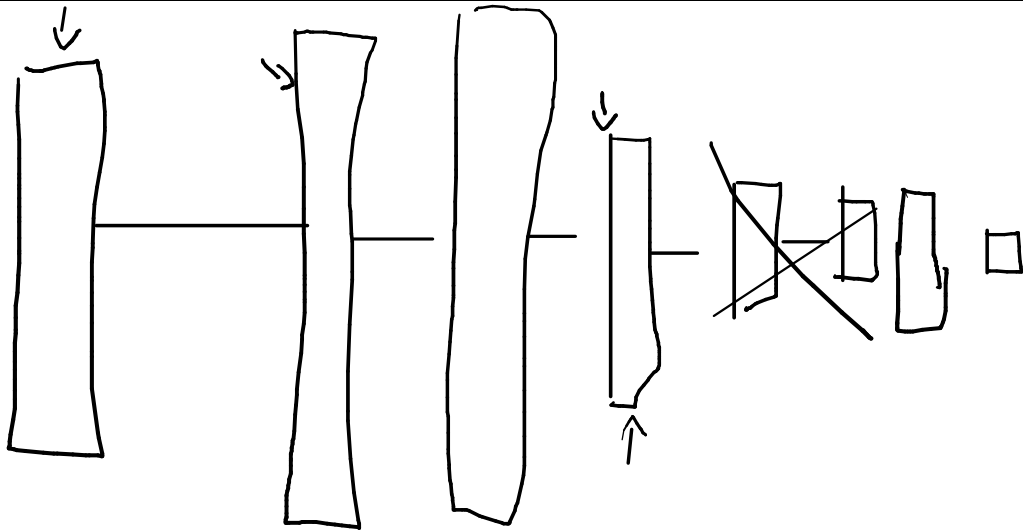
# Notes



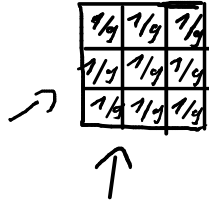
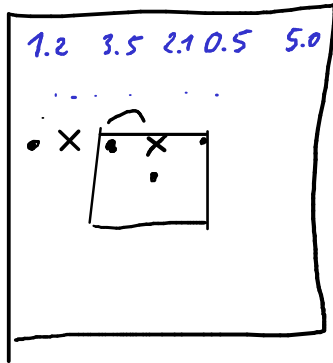
# Notes



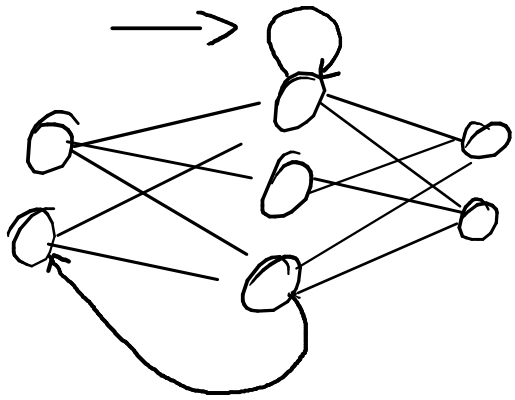
# Notes

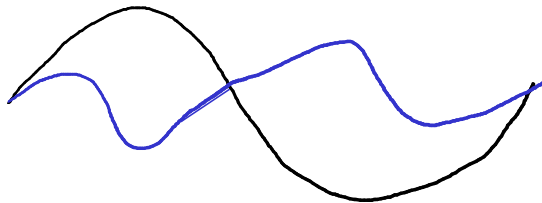


# Notes



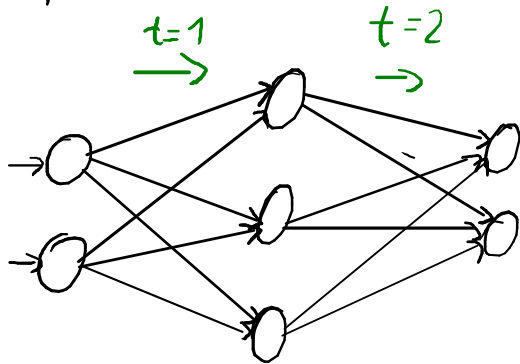
# Notes



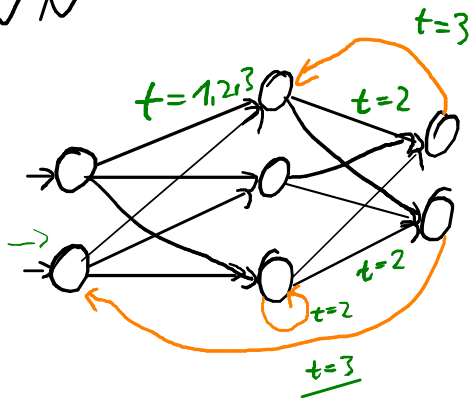


2.1

Feed forward NN

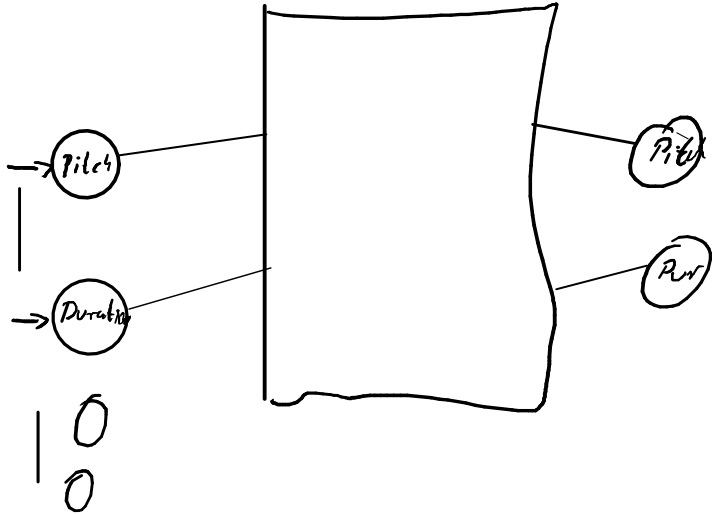


RNN

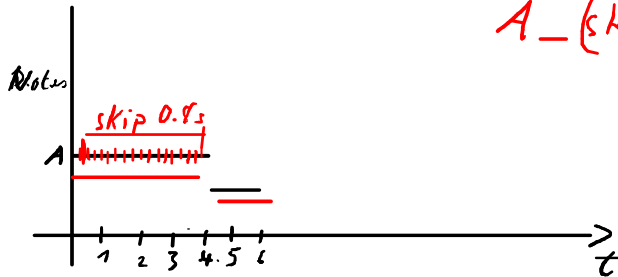




# Notes



# Notes



$A - (\text{skip } 0.8s) - C$

$A B B B B A B$

Arrows indicate transitions between the letters:  $A \rightarrow B \rightarrow B \rightarrow B \rightarrow B \rightarrow A \rightarrow B$ .

$P(B|B)$

